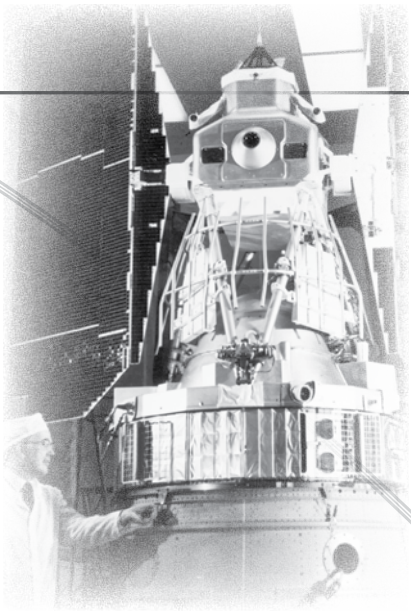
CHAPTER 6



ERA OF CHANGE

NASA's annual budget was $330 million in 1959. Just six years later, it had ballooned to $5.25 *billion*.[1] Over the next seven years, however, even as the space agency was putting 12 men on the Moon and busy pushing the envelope in launching a plethora of new generation of space probes, science satellites and application spacecraft, it saw its funding gradually being cut. By 1974, it had bottomed out at $3 billion.[2] In the FY 1973 NASA budget hearings, Gerald M. Truszynski, Associate Administrator for the OTDA, announced plans by the space agency to merge the STADAN and the MSFN into a single, more streamlined network.

Networks were developed under a certain sense of urgency in the early years of the space program. The need to respond to the Soviet Union and to put the American space effort on the fast track sometimes took priority over such matters as coordination of effort and minimizing of cost. It was, after all, a time of pioneering work with many unknowns. By the early 1970s, while the major emphasis on meeting program requirements had not diminished, coordination of these requirements, economic efficiency, and tighter management controls were being given a much higher priority. This fundamental shift to the pragmatic was felt—and felt hard—by those running its spaceflight tracking networks. As one NASA manager recalled, "There wasn't as much

money there, wasn't as much activity, [and we began] closing Apollo tracking stations, cleaning up after Apollo."[3]

It was in this atmosphere of renewed fiscal awareness that NASA merged the two networks to form what would be called the Spaceflight Tracking and Data Network, or STDN. There were also other reasons besides budget for a network consolidation. With the decline in scheduled human space activities after Apollo, the argument for a separate, manned-flight network became less compelling. For the engineers, technicians, managers and even the astronauts—the very men and women who had just put Americans on the Moon—there was a definite sense of let-down when it dawned on them that what seemed like an adventure which had just begun was now suddenly over. Bill Watson, the Program Executive at Headquarters who today oversees NASA's Ground Network, was fresh out of school and just starting his career at the time. Reflecting back, Watson said:

> There was a sense of what's next, what we should do next after Skylab and Apollo-Soyuz. It was hard for guys to get excited about scientific, robotic satellites to the extent that they were excited about the manned flights. There was a large hiatus there until the Shuttle program came along, and . . . a lot of folks left the program during that gap.[4]

The numbers reflect this. In 1970, the Agency's fulltime, civilian workforce stood at 31,223. By 1979, this had dropped to 22,633, a reduction of almost 30 percent. NASA cut its workforce by 7 percent in 1972 alone.[5]

Besides fiscal constraint and the rescoping of the Agency's mission, there were also good technical reasons for merging the networks. Both the STADAN and the MSFN were growing increasingly sophisticated. The clear separation of crewed versus uncrewed requirements that had so differentiated the two were becoming more nebulous due to the increasing number of high eccentricity (highly elliptical), high apogee observation satellites being launched. This new class of satellites had much in common from a tracking standpoint with an Apollo spacecraft traveling to and from the Moon. Meanwhile, network managers at Goddard thought that implementing the USB concept throughout the STADAN could serve as the common bond needed for a single, overarching, near-Earth network. All these factors served to provide Truszynski and his office with good reasons to merge the capabilities of the two networks. NASA's thinking was that, with a leaner network, fewer stations could actually provide a more flexible capability to support its upcoming workload for all near-Earth missions, both robotic and piloted.[6]

Network engineers understood that the existing geographical distribution of the stations could effectively be modified into a configuration that would be able to handle the total mix of missions which NASA at the time

foresaw for the latter half of the 1970s. Before this transition, there were 25 stations (19 STADAN and 6 NASA-owned, primary MSFN sites) spread over five continents (see maps in Appendix 1). The continual operation and maintenance of so many stations were, not surprisingly, expensive and required a great deal of manpower.

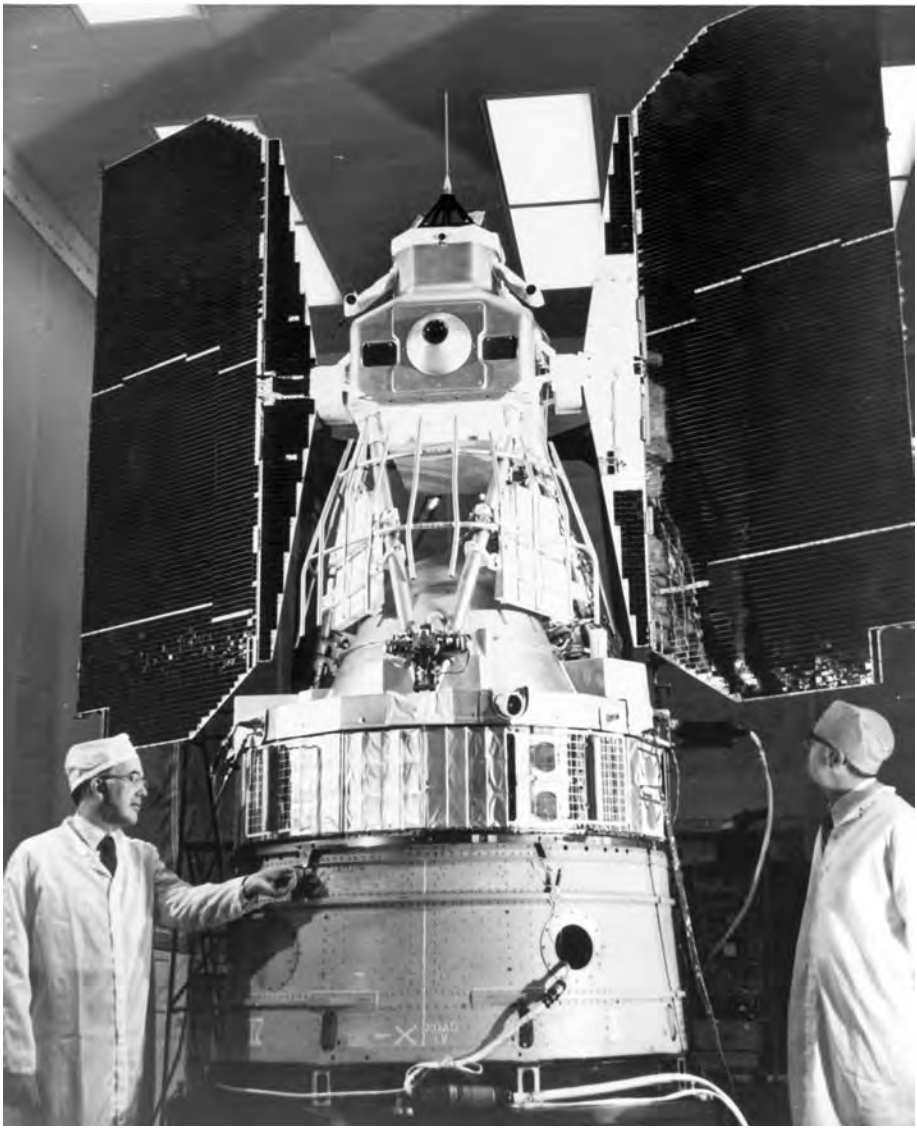
As of the mid-1960s, satellites were carrying much more powerful beacons so that telemetry—and not tracking—was now the pacing item. Technology was also advancing such that having fewer but better instrumented stations was now possible. Bill Wood, at the time Chief of the Manned Flight Operations Division and later Associate Director for Network Operations at Goddard, said of the change:

> As the Apollo program began to wind-down, we realized that both manned and unmanned tracking functions had to be consolidated. It simply became too impractical and too costly to maintain separate networks. . . . This was the time to change from the old to prepare for the future—Skylab, Apollo-Soyuz, Space Shuttle and the many unmanned missions also being planned.[7]

At the Directorate level of Goddard, the organization was reworked starting in January of 1971. Ozzie Covington now consolidated all network activities under him including the field stations, the Network Operations Control Center (NOCC), and communications. Under Jack Mengel were all the Project Operation Control Centers, data processing, and the large computers at the Center.

To implement the change, Goddard made sure that several requirements, both new and old, were going to be met. First, the high data rate, real-time TT&C capability of the manned network were retained since they matched well with the increasingly more complex satellite requirements that were then coming online. Many of the satellites were, in fact, approaching the complexity of and taking on the characteristics of human missions in terms of requirements for command and control, downlink data rates, and the higher operating frequencies at the S-band. Foremost among this new generation were "mega" satellites such as the Earth Resource Technology remote sensing satellite (ERTS), the High Energy Astronomy Observers (HEAO), and the International Ultraviolet Explorer (IUE). On the IUE, for example, the onboard telescope had to be moved at regular intervals by means of ground commands emanating from the GSFC. In general, telemetry rates were pushing state-of-the-art capabilities at 150,000 bits per second.[8]

When ERTS-A was launched on 23 July 1972, it was actually supported by the MSFN. Thus, there was an increased need in the unmanned spacecraft community for the type of technical capabilities which already existed in the MSFN. By 1974, work was well underway to modify the telem-



Technicians check out Earth Resource Technology Satellite ERTS-A at the General Electric Company Astro-Space Division in Princeton, New Jersey in 1972. Launched atop a Delta 900 launch vehicle from Vandenberg Air Force Base in July of that year, ERTS-A was the first in the series of Landsat remote sensing satellites, one of the most successful Earth resource application programs ever. Downlinking its data to the Goddard Space Flight Center at a rate of 15 megabits-per-second, ERTS-A was designed to last one year but was not deactivated until 2 January 1978. (Photo courtesy of the United States Geological Survey)

etry and command processing systems at the existing MSFN sites for compatibility to support science spacecraft. At the same time, however, NASA still had many of the less complex spacecraft such as the old Explorer series, which was still returning a healthy amount of data. These were generally the smaller, spin-stabilized satellites which could not accommodate the newer and larger, high-gain, directional antennas, and therefore, still had to operate at the lower VHF frequencies.

With this wide spectrum of needs, NASA required the full range of capabilities offered by both networks. Apollo just came to an end; the time was right for such a merger. By the end of 1974, the number of ground stations (STADAN and MSFN) had dropped to 17. Two years later, it went to 15. Of the eight ARIA, only four were now available for NASA support, the others having reverted back to the Air Force full time. Four of the five AIS were retired in 1969, leaving the *Vanguard* as the only network vessel to remain in service (it too retired in 1978).[9] Indeed, once the transition to the newly organized STDN was complete in 1976, network operations quickly became more standardized. NASA began to see greater returns from the slimmer network, all the while reducing the manpower needed for operations, logistics, and most importantly, cost.

While NASA did not present the consolidation of the STADAN and the MSFN to Congress until 1973, phase-down activities had already been taking place for some time. The first round of phase outs involved the STADAN stations at Blossom Point, Maryland; East Grand Forks, Minnesota; and Woomera, Australia in 1966. This was followed by shutdown of the temporary sites at Darwin and Cooby Creek in Australia; Lima, Peru (transferred to that country's university); and at Mojave, California in 1969 (the remnant of the old San Diego Minitrack station which had moved to Goldstone). A year later, St. John's, Newfoundland, on the eastern-most point of Canada, was shut down for good as was Fort Myers, Florida in 1972. By the time the STDN consolidation occurred, STADAN had, in fact, already streamlined down to nine stations (plus the NTTF in Greenbelt).

On the MSFN side, downsizing began soon after Apollo 11 when requirements for Apollo were carefully reevaluated by Headquarter's Office of Manned Space Flight. With little fanfare, NASA soon began reducing the number of MSFN stations as well, beginning with the shutdown of Antigua in the South Caribbean Sea on 15 August 1970. The Agency had determined that limitations on launch azimuth angles for flights following Apollo 13 would not require data from Antigua and that no increase in risk to mission success would be incurred as a result of the shutdown. In the words of a NASA spokesman, Antigua was simply the victim of "reduced requirements for NASA's worldwide tracking system."[10]

The station had a 9-meter (30-foot) USB system as its centerpiece. After Apollo 11, it was almost immediately relegated to a caretaker

status with the 17 Bendix employees and 11 Antiguans put on standby status.[11] Most of the equipment was transferred to other facilities in the MSFN. Although human spaceflight requirements for Antigua were soon deleted all together, the requirement to support other NASA launches out of the KSC still remained and the station stayed open at a reduced level. But the writing was on the wall. Soon thereafter, NASA pulled out of Antigua. The Air Force Eastern Test Range station on Antigua agreed to provide services to NASA as needed—on a cost reimbursable basis.[12]

The review board also showed that either the Corpus Christi Station in Texas or the Guaymas Station in Mexico could be closed. Had all factors basically been equal (including politics), the decision would have come down to fiscal considerations; that is, which one would yield the most cost savings. But Texas had one thing going for it that Guaymas did not. Due to its desirable location to support Earth resource satellites, it was the logical choice to remain open. By utilizing USB equipment from Guaymas, the station would be able to support both crewed and uncrewed programs. Because of this, the decision was made to keep Texas operational and close Guaymas. A meeting was held in Mexico City on 16 June 1970 with Mexican space officials, the U.S. Ambassador, and Gerald Truszynski discussing plans on how best to phase out the station. This was followed by a second meeting two months later in which it was agreed upon that NASA would remove two of the three major station systems for relocation to other parts of the network. The third system would be left in Guaymas to support Mexican space activities and programs of mutual interest to the Mexican science community and the United States.[13]

This was a good way to close a station. In addition to promoting goodwill between the two neighboring governments, the Mexican National Commission for Outer Space (CNEE) and NASA were, at the time, cooperating on two scientific projects. One was to develop a system using weather data acquired from U.S. satellites by using automatic picture transmission equipment. The other was to develop capabilities and applications for Earth observations using advanced airborne remote sensing instruments. The two countries were also completing plans for a cooperative project involving meteorological sounding rockets. After details of the agreement were ironed out, joint press announcements officially closing Guaymas were released by both governments on 12 November 1970.

By the following February, NASA's withdrawal from the station was complete. This brought to an end a decade of association during which America blazed a pioneering trail into space. From John Glenn's first flight into orbit to Apollo 11, Guaymas was there. Commenting on the legacy of the station, Dr. George M. Low, then Acting NASA Administrator, noted most fondly that the "cooperative establishment and operation of the station over the 10 most exciting years in space exploration stood as a tribute to the

friendship and understanding between the two countries."[14] In particular, he singled out members of the Mexico-U.S. Commission for Space Observations who first laid the groundwork in 1959 to make Guaymas possible, specifically recognizing: Hugh Dryden, Chris Kraft (Director of Flight Operations at the Manned Spacecraft Center), Ralph Cushman (Special Assistant, Office of the Administrator), and Dr. Eugenio Mendez Docurro (Secretary of Communications and Transport, Av. Universidad Xola). This was quite the fitting tribute to a decade which saw the sleepy little railroad town of Empalme, Sonora (12 miles outside the actual city of Guaymas) thrusted into the international space forefront to become, even today over 30 years later, a source of pride for the Mexican people.

As the transition took place, plans regarding which stations to keep and which to close could change quickly, and often did. Take Canary Island, for instance. In the summer of 1973, NASA Headquarters proposed a five-year extension to the Spanish government that the station be kept open until 1978, when NASA's TDRSS was then scheduled to become operational.[15] The station seemed safe for another five years. Several requirements still needed support including telemetry reception from the Apollo Lunar Surface Experiment Packages that had been left on the Moon by the astronauts and the upcoming Apollo-Soyuz Test Project that would take place in 1975. Ironically though, it was this same requirement to support ASTP that ended up providing the impetus needed to shut down Canary Island.

This twist of fate came about due to the requirement for live television, a critical requirement on the highly publicized ASTP. It had been anticipated (correctly) by NASA that this particular mission, as the first international human spaceflight between the two Cold War rivals, would draw worldwide interest not seen since that of Apollo 11 five years earlier. As early planning requirements for extensive real-time coverage were being developed (jointly by the ASTP Program Office in Washington DC and Moscow), it became apparent to both that this requirement was not going to be met effectively using existing MSFN capabilities. Something better was needed. The Agency would use the ATS-6 to directly receive television signals from the Apollo spacecraft and then retransmit them to a ground station in Spain, rather than depending on the ground stations alone.

Fallout from this decision on Canary Island came quickly. On 22 January 1975, Truszynski sent a letter to the Director General of the Madrid Station (of which the Canary Island station was a part of) that NASA has "regretfully come to the conclusion that both near and long term data acquisition requirements do not support the continuation of the Canary Island station and would desire to close the station as soon as possible."[16] Canary Island's fate was officially sealed two days later by a notification from Truszynski to NASA's Assistant Administrator for International Affairs that services on Canary Island were no longer needed and that the State Department was

requested to take appropriate actions as soon as possible to shut down the station. Thus, Canary Island went from being a crucial land station in the eastern Atlantic to "not necessary" in the span of not even a single mission. In a way, it was a harbinger of things to come as ATS-6 tested out the new concept of space communications, one that would rely almost exclusively on space-based satellites to do the job that ground stations once did.

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Soon after Apollo 11, the Guam and Hawaii stations took center stage in a budget fight between the Bureau of the Budget and NASA. During the FY 1971 budget process, the Budget Bureau notified NASA that Guam and Hawaii were going to be phased out and their operations transferred to the DOD satellite control station on their respective island. Each year, with a few exceptions, every department and agency of the federal government has to negotiate the "necessary evil" of the budget process; NASA was no exception. While budget negotiations were an annual ritual, what the Bureau was telling NASA in this instance was considered by the space agency as being somewhat "out-of-line." The Budget Bureau's position was that NASA should shut down these stations, but that in order to "alleviate to the extent possible impact on mission support," the DOD would "give the NASA manned missions highest priority in workload allocation."[17]

In November 1969, Administrator Thomas Paine rejected this proposal outright, making it clear that this was indeed an assumption of fait accompli, one not based on any DOD-NASA discussion after it was proposed at the start of the budget process. A paper was drafted explaining why NASA believed that any such consolidation would be neither operationally feasible nor cost effective. NASA's viewpoint was based in part on a preliminary joint NASA-DOD sponsored study to evaluate the merit of consolidating the NASA and DOD network facilities on Guam and Hawaii. No long-term operational costs were identified which would have offset the substantial immediate cost of modifying and relocating the equipment and expansion of facilities required to handle the high-priority functions of both agencies. Before sending this paper to the Budget Bureau, Paine confirmed that "responsible officials in the Air Force agree with us that the conclusions of this study are still valid."[18]

This did not end the matter however. Three months later, the Bureau of the Budget once again informed NASA that the Guam and Hawaii stations were to be phased out. This time, in a strongly worded letter to Robert P. Mayo, Director of the Bureau of the Budget, Paine voiced the Agency's concern that they now appeared to be under direction, without prior consultation, to take an action which was operationally and economically unsound in the view of both NASA and the U.S. Air Force. Since he was at the time accompanying the Apollo 11 crew in the "Giant Leap" victory tour in the Far



Gerald M. Truszynski (far left) rose through the ranks to become NASA's Associate Administrator for Tracking and Data Acquisition from 1968–1978. This picture shows Truszynski when he was Head of the Instrumentation Division participating in the 27 January 1953 ground breaking ceremony of the NACA High-Speed Flight Research Station (which became the Dryden Flight Research Center) on the northwest edge of Rogers Dry Lake in the Mojave Desert. Pictured with Truszynski were Joseph Vensel, Head of the Operations Branch; Walt Williams, Head of the Station, scooping the first shovelful of dirt; Marion Kent, Head of Personnel; and California state official Arthur Samet. (NASA Image Number E-980)

East, Paine volunteered to change his travel plans so that he could personally look into the situation at the NASA and Air Force stations in Hawaii on his return. In the meantime, he directed Truszynski and his office to review again the requirements on both islands with DOD officials. Drawing a line in the sand, Paine concluded his letter to Mayo in no uncertain terms, saying, "Unless new information is developed in my visit or in the review, I will then formally reopen this matter with you, and if necessary, the President."[19]

Guam and Hawaii went on to survive that year's budget process. In fact, both stations went on to become among the longest-serving STDN sites, remaining operational for another 19 years, finally closing in 1989.

As NASA stations began to close around the globe, none was more of a political target than the Johannesburg Station in the Republic of South Africa. This was one of the few communication complexes where the DSN and the STADAN shared a location. Roots of the DSN go back to the late 1950s. As the United States moved from the realm of Earth-orbiting satellites to begin sending probes to the Moon and beyond, a "World Net" was established by the DOD's Advanced Research Projects Office. This World Net formed the nucleus of what would go on to become the DSN. In order to maintain continuous coverage of space probes departing the planet as Earth rotates, three sites are needed, each situated about 120° apart. The DOD—and later NASA—had placed the first two sites at Goldstone, California and Woomera, Australia. Completing the World Net was the construction of a third station in the country of South Africa, where a government-owned, 4,000 acre grassland valley near the Hartebeestpoort Dam 65 kilometers (40 miles) north of Johannesburg was provided.[20] The station became operational in June of 1961.

To meet tracking requirements in the Southern Hemisphere, a $5 million expansion at the Johannesburg complex was done three years later that brought the number of stations to three. One was run by the U.S. Air Force to control its satellites. Due to its obvious military nature, the station was staffed entirely with Americans. The other was a NASA satellite tracking station. The remaining site, for all intents and purposes, was part of the NASA station but was operated for the Smithsonian Institute, its roots dating back to the IGY and Minitrack.[21] Unlike the Air Force, NASA staffed these two stations with South African workers and normally only had a U.S. liaison officer present onsite. Under a 1960 agreement with the space agency, the National Institute for Telecommunications Research, a part of South Africa's Council for Scientific and Industrial Research (CSIR), had full responsibility for management of the station which they operated so as to meet NASA's technical requirements. The station was fairly extensive. At its peak, the NASA side of Johannesburg employed some 280 South Africans of whom about one in five were black.[22]

Even as NASA began working with the South African government to establish stations there, the potential fallout from that country's racial segregationist policies was not lost on many in the United States. NASA was fully aware that an agreement with a government espousing such policies could become a political flashpoint. But at the same time, it could not just discard the technical merit of such a location. Here's why: for optimal coverage of interplanetary probes launched on trajectories from Florida, an antenna was best placed as far south as possible, preferably deep in the Southern Hemisphere. The Republic of South Africa, being on the very southern tip of the conti-

nent, was ideal. As unfavorable as the South African political climate was, it was actually the most democratic and most stable government accessible to the United States on the continent at the time. To keep its options open, even as negotiations were being held with South Africa, NASA still looked at other locations, particularly those in southern Europe. These, in order of preference, were Sicily, Sardinia, south Spain, and south Portugal. Headquarters also looked into a possible cooperative arrangement with France, which at the time was considering the purchase of a 26-meter (85-foot) antenna from the Collins Radio Company to build a ground station of its own on the Normandy peninsula.[23]

These were more than just cursory looks. Site survey teams consisting of members from Headquarters and the Jet Propulsion Laboratory were sent to all these locations as NASA wrestled with whether or not to proceed with South Africa. In the end, it was decided that the geographical location, along with the country's already robust scientific community and the expressed enthusiasm of the South Africans, best advocated putting stations there.

It did not take long for tensions to arise. Accusations centering on the station started to surface even back in 1962, that South Africa might be putting pressure on the U.S. government for NASA to adopt a segregationist policy there. This was a serious concern, so much so that Associate Administrator Edmond Buckley wanted an early evaluation of the matter by asking the State Department to look into the situation.[24] Time did not assuage the tension between the two governments, though. In fact, things only got worse. The situation came to an early head when in May 1965, the United States asked South Africa for permission to have a squadron of advance-planes from the aircraft carrier USS *Independence* land at airports when the ship was scheduled to dock at Capetown. The government granted the Americans permission, provided the planes' crews were white. Up until then, American planes had often landed at South Africa airports and on occasion, there had been mixed-race crews including blacks. However, never before had the South Africans explicitly asked for all-white crews.[25] This caught the State Department totally off guard. In an attempt to clarify the meaning of the South African response, the United States asked if this was a condition or a suggestion. If it was a condition, South Africa was told it would not be accepted. If it was a suggestion, no guarantees could be given. With no clear response from the South African government and not wanting to escalate the already well publicized series of events, the USS *Independence*, in the end, skirted the issue by bypassing Capetown altogether.

American resolve was further tested just a month later when, for the first time, pressure to actually shut down the station officially came from the South African government. This time, Premier Hendrik F. Verwoerd announced in a press release that he had told the United States it cannot employ "negro scientists in the South African stations," and that his govern-

ment "would not admit American negroes if they were assigned to work at the tracking stations."[26] Verwoerd's comments on the tracking station staff seemed to most observers at the time to have been a condition, deliberately made, so as to provoke an American response. An opinion editorial came out that same week in the *South African Sunday Times* declaring that the United States would have to decide whether it can "afford morally" to overlook Dr. Verwoerd's remarks. The irony was that just three years earlier when South Africa's role in NASA's tracking network was being heralded, the same newspaper headlined "South Africa has Important Part in U.S. Moon-Shot."[27]

On the other side of the Atlantic, the station became a major target of blacks and liberal politicians who protested that the United States should not be putting money into a country with whose racial policies we do not agree. Into the 1970s, numerous congressional inquiries and hearings before the House Subcommittee on Aeronautics and Space Technology were conducted. Led mainly by prominent liberal members of the Democratic Party, these hearings aimed to determine just what exactly NASA was doing in South Africa. To that end, they looked at what the United States was doing to improve the working and living conditions and opportunities for black South Africans employed at Johannesburg. NASA administrators from Headquarters also answered questions before the House Committee on Foreign Affairs regarding the specific racial breakdown of employees, salary breakdown by race, wage practices, and NASA's hiring practices of Black Africans. (The irony was that NASA did not do any hiring in South Africa. CSIR hired African employees from an agricultural group resident in the area of the station while whites were hired through normal CSIR employment channels for technical assignment.)[28]

As hearings progressed through the mid-1960s to the mid-1970s, the issue intensified to the point where heightened scrutiny was placed on even the smallest of details, such as educational assistance, Christmas bonuses paid to whites versus negros, eating facilities and provisions for medical services. On one side of the aisle, members of the House Congressional Black Caucus, led by Representative Charles B. Rangel of New York, viewed the station as an egregious symbol of American acquiescence to apartheid. Others in Congress, led by Representative Olin E. Teague of Texas, Chairman of the House Space Committee, argued that the station was really South African, not American, since NASA did not employ any Americans in South Africa. Information gathered by NASA at the behest of Representative Charles C. Diggs of Michigan showed that, whereas blacks held about 25 percent of the jobs at the station, they received only about 5 percent of the wages paid by NASA through CSIR. In 1972, after returning from a visit to the site, Diggs reported that black employees were barred not only from the station cafeteria but from most of the technical and all of the supervisory jobs and from the technical training programs. Representative Rangel charged that gross disparities existed between fringe benefits given to white and black employ-

ees, benefits such as sick leave, vacation time, and medical benefits. To support his case, Rangel presented numbers showing that the highest paid black employee—a "skilled laboratory assistant"—earned $2,005 per year, just barely more than the lowest paid white employee—a "raw trainee"—who earned $1,930 a year.[29]

Even when there was good news for NASA regarding South Africa, it was tainted by what could only be called handwriting on the wall. In May of 1973, a House bill that would have cut $3 million of NASA funding for stations in South Africa was defeated. However, in defeat, more votes than ever before (104 to 294) were rallied. That same month, Massachusetts Senator Edward M. Kennedy introduced an amendment to cut off funds for the station, an amendment he later withdrew but only after the Senate Space Committee's new chairman, Utah's Frank E. Moss, promised to look fully into the matter in the fall session. According to Moss, the unconditional shutdown of the station would have meant that another station would have to immediately be established elsewhere. If this had to be done, the replacement cost would have amounted to around $35 million, something that would have been difficult to justify on the bill that late in the budget process.[30]

Throughout this debate, NASA consistently countered that local improvement programs which accompanied the stations were in fact making a difference. For example, the United States was, at the time, providing approximately $109,000 a year (1973 dollars) on improvement programs for the black station community. Among them was the building of houses for the African staff, at the rate of one completed every two months, and the construction of an elementary school. By 1974, 18 new houses had been constructed plus the school. Under the agreement between NASA and CSIR, the South Africans provided the initial construction funds which were then reimbursed by the United States upon completion. NASA also operated a small medical facility onsite, the services of which were made available to the Black African staff and their families. Although it was only staffed part time—a nurse was on duty three days a week and a doctor visited once a week—it was, nevertheless, one of the very few modern medical facilities in the Hartebeestpoort area that provided services to the black community, and as such, was well used. However, station critics in Congress regarded these improvements as merely cosmetic, noting that South Africa seemed not to think the station important enough to its own interests to justify making exceptions to the rules of apartheid. "The system is so unyielding," said an aide to Charles Rangel, "that if the U.S. had forced the point, South Africa would have just kicked the station out."[31]

As things turned out, Senator Kennedy did not have to wait until the fall session. After more than a decade of defending the station, on 10 July 1973, Administrator James C. Fletcher announced that it would begin pulling out of South Africa the following summer and would withdraw U.S. support entirely by late 1975. The phase-out would be done in two stages, starting first with the

DSN side in June of 1974 followed by the STADAN side after completion of the near-Earth phase of the Viking Mars missions.[32] The decision to phase out Johannesburg did not, however, signal the immediate cessation of all NASA activities in South Africa, just its tracking stations. Meteorological data collection as part of NASA's worldwide program to conduct high altitude air sampling in all hemispheres continued. Data analysis for the LANDSAT-2 satellite (in which the U.S. was one of roughly 50 countries involved) and lunar sample analysis continued for years thereafter, some even to this day.

As NASA pulled out of Johannesburg—and other stations for that matter—what to do with the equipment and hardware usually came down to two options: 1); Remove all or part of it at the Agency's own expense, with the implied, parallel responsibility to restore the site to its original condition; or 2); Dispose of the property, all or part of it, within the host country in accordance with arrangements agreed to beforehand by the two governments. In South Africa, the cost to dismantle the Deep Space portion (DSS-51) would have amounted to $643,500 with an additional storage cost of $11,060 (1975 dollars).[33] Based on this estimate, NASA determined that its real property interests at the tracking station constituted foreign excess property which had essentially no commercial value. Eventually, it was concluded to be in the best interest of the U.S. government to either donate or abandon in place the property to the South Africans. In doing so, it was mutually understood that the assets would be relinquished with the provision that no further U.S. obligation or liability remained. NASA, in essence, washed its hands of South Africa.[34]

Nevertheless, finger-pointing continued. Noting that the Agency had previously closed down two similar tracking stations in just the past year—Fort Myers, Florida, and Woomera, Australia—the Agency said that the South Africa decision was based entirely on technical requirements and was in no way a response to political pressure. Critics in Congress disagreed. "Frankly," said a spokesman for Senator Moss, "I think they just saw the handwriting on the wall, the message being that the station was becoming an embarrassment." Moss himself later released a statement praising NASA for its decision to pull out, adding "Apartheid has always been repugnant to me."[35]

In reality, NASA began planning phase out activities for the station as early as 1971. Its official position was that there would be an absence of requirements for long-period, near-Earth, Southern Hemisphere coverage after Viking left for Mars in 1975. Following that, deep space requirements could be handled by the DSN stations at Canberra, Goldstone, and Madrid. With this plan in mind, discussions were held with CSIR in August of that year to give them as much time as possible to work out staffing plans. A concern at the time for both countries' space programs was to not just abandon the station but rather, retain enough competent staff through the transition period as it moved from being a jointly sponsored site to one that was fully South African.

In the phase out discussions with CSIR, the fate and future of the Black African staff were, in fact, discussed at length, down to the number of Black Africans which might remain employed after NASA relinquished funding. There was particular concern on NASA's part that Black African staff would be declared "redundant" and whether they would be treated equitably relative to the white staff. At Headquarters, Gerald Truszynski, in his discussions with Dr. Frank Hewitt of CSIR, felt that the South Africans appreciated the American position, with Hewitt saying he "reflected a genuine concern for the future of this group."[36]

A legacy of these discussions was that it led the South Africans to implement several policy changes with regards to Black station staff members. One had to do with the pension they were receiving. At the time, the Black African staff members were covered by a different benefits plan that was generally (and obviously) inferior to that of the white staff. This "Provident Fund Plan" was soon changed so that the same formula was used in calculating the pension for all staff members. In addition, after these changes were made, CSIR allowed the Black African staff who were declared "redundant" to, where appropriate, continue occupying their houses, thereby enabling them to look for other employment before moving their families off the station site. Arrangements were also made for CSIR to provide a vigorous outplacement service and reemployment counseling. On the other hand, the one service which the South Africans did not continue after NASA ceased its funding was the secondary school bursary program which the Agency had started. CSIR deemed this to be outside of their normal responsibility and charter as they had many Blacks employed in their agency's other activities who were not receiving any educational assistance.

In the end, two-thirds of the Black staff (39 out of 59) were released after NASA pulled out of Johannesburg.[37]

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As a principal site in the Southern Hemisphere, Tananarive (TAN) had been busy, supporting a host of science satellites as well as all the Gemini, Apollo and Skylab missions. The routine began to change in 1972 when the Malagasy government underwent a series of political upheavals. In May of that year, the president of the ruling Social Democratic Party, which had been in power since Madagascar first gained independence from France in 1960, resigned under political pressure. The unrest continued over the next three years, culminating with the brutal assassination of the military dictator which put the country under martial law in February of 1975. Before long, a new Marxist regime was formed under the leadership of a 38-year-old revolutionary named Didier Ratsiraka. Under President Ratsiraka, known in the region as the "Red Admiral," the government became highly centralized and com-

mitted to revolutionary socialistic ideals. (These policies did not change until the 1990s only after the formation of new political parties.)

One of the first foreign policy changes that Ratsiraka made was to impose a rent on the United States for operating a NASA ground station in his country. In the original memorandum 20 years earlier which both countries signed establishing a site near Majunga, it was agreed upon that there would be no exchange of funds and no rent exacted for use of land. But now, Ratsiraka was demanding $1 million per year, *retroactive to 1963* on back taxes. This was a demand that the United States obviously could not agree to.

Negotiations were conducted but to no avail. A few weeks later, the Supreme Council of Revolution of the Malagasy Republic forced the station closed. This action came upon NASA unexpectedly. During that time, GSFC was still improving on the station and in the process of adding a Unified S-band antenna. Under the guise of avoiding "possible maneuvers of sabotage," President Ratsiraka immediately placed it under military control. The Station Director and Bendix workers with their families were allowed to evacuate, but all equipment had to be left behind. At the time of closure, there were the two Goddard appointed NASA employees and 50 Bendix workers, along with their dependents—148 rather apprehensive Americans total—at the station.[38]

With the abrupt shut down, Goddard had to make some quick changes in order that support for Apollo-Soyuz, which was to launch the very next day, would not be disrupted. They improvised by tasking the geosynchronous ATS-6 to serve as a data link. Workload from other satellites was shifted to other stations. ARIA instrumentation aircraft and the *Vanguard* were repositioned to help support other launch activities out of the Eastern and Western Test Ranges. These changes resulted in some temporary scheduling problems but otherwise proved adequate and Apollo-Soyuz went on to be an unparalleled success.

Over the course of the next five years, the Malagasy government periodically allowed NASA back into the country to remove equipment. On 3 April 1980, the last of the remaining hardware that NASA still wanted was removed from Tananarive. By diplomatic note, the remaining U.S. property was turned over to Madagascar the next day. This note, which was actually received by the U.S. Embassy the previous October, expressed essentially an agreement on the list of equipment NASA would remove and the monetary settlement. The removal process, in effect, was the final act that brought to an end five years of negotiations by the State Department to repatriate NASA equipment following the forced closure of the station.[39] Besides the stress and disruption experienced by the staff and their families, the closure also had an effect on NASA in terms of operating cost. After Tananarive was shut down, the *Vanguard* was called on to fulfill some of its requirements. In the mid-1970s, the annual cost to operate a tracking ship was quite high, about $6

million per year. By contrast, a land station like Tananarive cost around $2.8 million, or less than half that of a ship.[40]

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One of the first actions in the reorganization for STDN took place at the Goddard Space Flight Center where management of both networks was consolidated as early as May of 1971. Two new directorates, the Mission and Data Operations (M&DO) Directorate and the Networks Directorate (ND), replaced the Tracking and Data Systems (T&DS) and Manned Flight Support (MFS) Directorates. In this new arrangement, divisions within the M&DO managed the data processing activities and the computing requirements of the network. The ND became responsible for operation of all the STDN elements, from NASCOM to the ground stations and the satellites. In a harbinger of things to come in the 1990s, it was at this time that the Networks Directorate formed the Network Office for International Operations, which allowed GSFC to start handling some of the foreign policy work that up until then had rested exclusively in the domain of NASA Headquarters in Washington, DC. This move seemingly made sense at the time, coinciding with preparations for Apollo-Soyuz which demanded a lot of technical interaction with Soviet Union network engineers at the working level.

Although STDN was considered a new network (or at least a greatly retooled old network), much of the way in which it was run continued as before, including usage of acronyms—a well known hallmark of NASA. The MCC in Houston continued to serve as the focal point during human space missions with the responsibility of directing all ground stations when a flight was in progress. Meanwhile, the NOCC (Network Operations Control Center) in Greenbelt continued in its role of controlling the network including overseeing all network preparations leading up to the launch of a human space mission.

In this capacity, NOCC engineers monitored console displays and established direct voice links amongst all the mission elements such as the launch site at the Cape, the network ground stations and the appropriate Project Operations Control Center (POCC). These POCCs that began emerging during the previous decade were essentially individual operation control centers at the GSFC that were built to specifically control certain types of satellites such as the Applications Technology Satellite (ATSOCC) or the Orbital Astronomical Observatory (OAOOCC). Not to be left out were the multimission "umbrella" centers such as the Multisatellite Operations Control Center (MSOCC) and the Mission Operations Control Center (MISSOC) that scheduled network support for all classes of satellites and assigned each station a weekly list of satellites that were to be monitored.

To accommodate the data downlink from the new generation of satellites that were now nothing short of orbiting laboratories, Goddard enhanced the STDN with increasing centralized capabilities. A typical change was the greater reliance on electronic data transfer methods in the late 1970s with the implementation of systems such as the Telemetry Online Processing System (TELOPS), that eliminated the need to ship magnetic tapes from the field stations. Raw data was, instead, transmitted over communication lines to a dedicated storage system at GSFC.

The POCC themselves also continued to acquire improved technology, permitting scientists stationed at Goddard to manipulate the orbits and attitudes of satellites with greater ease. Take for example, as a progenitor of touch-screen technology, a scientist seated at the IUE Operations Center. This person could, by simply pointing a light-pen at a specified portion of a video display, swing the IUE telescope around to look at another part of the



The Goddard Space Flight Center has been home to many Project Operations Control Centers over the years. Shown here is the Space Telescope Operations Center where commands to the Hubble Space Telescope (HST) originate and where its systems are monitored. The picture was taken in December 1999 as engineers monitored activities during the telescope's third repair mission. Today, command and control of the HST is done mostly from the Space Telescope Science Institute in nearby Baltimore. (Image courtesy of NASA, available at *http://hubblesite.org/gallery/spacecraft/01/)*

sky. In a move exemplifying NASA's continual effort to obtain ever better quality videos, scientists supporting Earth resource and remote sensing satellites also received a special Image Processing Facility (IPF) at Goddard that provided video and pictures that were continuously corrected for distortions introduced by spacecraft equipment or during transmission—a progenitor to today's high definition television transmissions.[41]

Stations reported back each week to the MISSOC on the performance of their satellite coverage. To measure performance, matrices were set up and grades given on how well stations were doing their jobs, whether excellent, good or poor. The success rate for each station was measured by how many passes were supported and how much low bit-error data was captured. This was then compared to how much *could have been* or *should have been* captured. The focus was primarily on the amount of data captured on satellite passes and not on cost of operations. This may not have been the best way to gauge how well a station did. Former Quito Station Director Charles Force said, "I thought at the time that was a mistake. You need to have some kind of a balance between how much you are spending and how well you are doing the job. But it was totally focused on how well the job was being done."[42]

The system was also not without its flaws. Force remembered an example that always puzzled him:

> It was my first exposure to performance evaluation using a matrix and what I learned from that personally, was that you have to be very certain the matrix measures what you want. One example I remember. If a satellite came over the horizon and the station was tracking it and the receiver failed, and they start losing data, they would then be scored down so [that] if they got the receiver up before the end of the pass and covered the balance of the pass, they got a lower score than if they didn't get the receiver back up.

In other words, the station would actually get penalized if it successfully recovered from the receiver failure than if it hadn't. "That was the system," said Force. "That was idiotic why it was set up that way. I have no idea because I was not in on the early days of the STADAN."[43]

While the original STADAN side of the house relied on these metrics to grade the performance of its stations, no such matrix was used on the MSFN side of the house. This apparent dichotomy in the way the two networks operated prior to their merger can, in large part, be traced back to the way the two networks came about—and the competition that followed. For years, while both the STADAN and the MSFN were run by Goddard, they were separate, up through the directorate level. Specifically, "Code 800" ran the MSFN while "Code 500" ran the STADAN.

From the onset, there were cultural differences, and with it, friction. Some would go so far as to call it jealousy. While Code 500 dated back to Jack Mengel and the team that created Minitrack, Code 800's heritage was basically an offshoot of Langley—a lot of the people actually came over to Goddard from Langley. Because of this, the two networks had different heritages and different cultures in everything from the way they operated to how people were used in the field. STADAN stations, for example, were generally more "remote" in the way they operated in the sense that data was gathered and sent back to Goddard where it was then assimilated and processed. Thus, there was a lot of effort by technicians and engineers physically at GSFC running the computers. Unlike today where desktop and notebook computers can be found in every office, this way of centralization made sense at a time when mainframe computers were required to do the massive calculations. These mainframes were expensive, to put it mildly, and required a fair amount of maintenance. Therefore, the STADAN had a lot higher percentage of its technical expertise stationed at Goddard in proportion to the field. This was exactly the opposite of the MSFN, that had more computational capability in the field and, therefore, had more of its share of expertise assigned to the field.

There has been the conception throughout the years that by nature of its mission, the MSFN was somehow more glamorous and had a higher profile (public exposure) than the STADAN, and therefore, got more attention and resources. This was, in all likelihood, exactly what happened. It was an undeniable fact that the MSFN received more attention than the STADAN in the one area where it most mattered: funding. In 1968, for example, two-thirds of the budget for Goddard's tracking operations went to the MFSN whereas one-third went to the STADAN.[44]

It's been said that where the money lies, so lies the priority. This apparent inequity was well recognized and unfortunately, resented within the STADAN system. An "us-and-them" attitude developed in many circles. As Force put it, "They didn't talk to each other that much."[45] The presence of this "sibling rivalry" is probably not too surprising considering the diversity of the people and their talent that was (and is) the Goddard family.

The years have shown that while such differences and strong feelings existed, they were worked out and the STDN moved on. The network that came out of it was a far better and more efficient network than before. Force would later say compellingly of the big picture, "The people did work together [and] the job was done successfully. There were an awful lot of good people that did work together and there was an awful lot done right and successful."[46] Indeed, the ensuing three decades have proven that.

By 1975, the merging of STADAN and MSFN was complete. Table 6-1 is a glance of the reorganized network in the mid-1970s.

Over time, these sites adjusted to the changing demands of the integrated network to support tracking of both human spaceflight missions

Table 6-1: Ground Stations of the Spaceflight Tracking and Data Network in 1975[47]

Station (Location)	Call Sign	Latitude Longitude	Year Established	Year Phased Out	Original Network	Primary Capabilities
North America						
Alaska (near Fairbanks)	ALASKA	64˚59'N 147˚31'W	1962	1984 (Transferred to NOAA)	STADAN	GRARR, MOTS*, SATAN 40, 45, and 85-ft dish antennas
Goldstone (Mojave Desert, California)	GDS	35˚20'N 116˚54'W	1967	1985 (Turned over to DSN and re-designated as DSS 16 & 17)	MSFN and STADAN	30 and 85-ft USB
Merritt Island (Kennedy Space Center, Florida)	MILA	28˚25'N 80˚40'W	1966	Still operating	MSFN	30-ft USB C-band radar
Network Test and Training Facility (Goddard Space Flight Center, Maryland)	NTTF	38˚59'N 76˚51'W	1966	1986 (Transferred to Wallops)	MSFN and STADAN	30 and 59-ft antennas
Rosman (North Carolina)	ROSMAN	35˚12'N 82˚52'W	1963	1981 (Transferred to DOD)	STADAN	Two 85-ft antennas GRARR; three SATANs, MOTS ATS telemetry and command
White Sands (New Mexico)	WHS	32˚21'N 106˚22'W	1961	Transformed into TDRSS ground terminals	MSFN	C-band radar VHF voice
Pacific						
Guam	GWM	13˚18'N 144˚44'E	1966	1989	MSFN	30-ft USB
Hawaii (Kokee Park, Kauai)	HAW	22˚07'N 157˚40'W	1961	1989	MSFN	Two yagi command 14-ft antennas C-band radar 30-ft USB

* While the MOTS cameras remained at the stations, they were no longer required for calibration after phaseout of Minitrack. They continued to be tracking devices for geodetic research, photographing the Pageos spacecraft in the mid-1960's. By the 1970's, MOTS was no longer an operational network.

continued on the next page

Station (Location)	Call Sign	Latitude Longitude	Year Established	Year Phased Out	Original Network	Primary Capabilities
South America						
Quito (Ecuador)	QUITO	37°00'S 78°35'W	1957	1982	MINI-TRACK	40-ft antenna SATAN, three Yagi command, MOTS
Santiago (Chile)	AGO	33°09'S 70°40'W	1957	1989	MINI-TRACK	40-ft antenna GRARR SATANs (2 receive, 1 command) Yagi command, MOTS
Atlantic						
Ascension Island	ACN	7°57'S 14°35'W	1967	1989	MSFN	30-ft USB (Also had a 30-ft DSN antenna which was phased out in 1969)
Bermuda	BDA	32°15'N 64°50'W	1961	1998	MSFN	C-band radar 30-ft USB
Europe						
Madrid (Spain)	MAD (RID after 1984)	40°27'N 4°10'W	1967	1985 (Transferred to DSN and re-designated as DSS 66)	MSFN	85-ft USB
Winkfield (England)	WNKFLD	51°27'N 00°42'W	1961	1981 (Turned over to the British)	STADAN	14-ft antenna SATAN, MOTS Yagi command
Australia						
Canberra (Honeysuckle Creek)	CAN	35°24'S 148°59'E	1966	1984 (Moved to Tidbinbilla, transferred to DSN and re-designated as DSS 46)	MSFN	85-ft USB
Orroral Valley	ORR	35°38'S 148°57'E	1965	1984 (Turned over to the University of Tasmania)	STADAN	85-ft antenna Two SATANs Yagi command MOTS



A "Spanish Watchman" on the hills overlooking NASA's Madrid Spaceflight Tracking and Data Network Station and its prominent 26 meter (85 foot) Unified S-band antenna. The station at Fresnedillas, some 50 kilometers(30 miles) west of the city of Madrid, was the NASA ground station that tracked *Eagle* to the lunar surface on the historic flight of Apollo 11. The station was phased out and transferred to the nearby Deep Space Network site at Robledo in 1985. (Photo courtesy of Larry Haug and Colin Mackellar.)

and applications satellites. From a purely technical standpoint, the augmentation of former STADAN stations with Unified S-band hardware was the biggest single improvement that NASA took to provide a common capability across the STDN stations. By the mid-1970s, this had been done at the Goldstone Apollo site (1972), Fairbanks (1974), Orroral Valley (1974), and Santiago (1974).

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How busy a station was depended on how many spacecraft it was assigned to track. It was not necessarily true that the largest and best-equipped stations were the busiest. It all depended on where a particular ground station was located. For example, it may not be surprising that Fairbanks, Rosman,

Canberra, and Goldstone—all home to one or more 26-meter (85-foot) antenna—were normally on four-shift, 24/7 operations. Contrast that with the Madrid Station at Fresnedillas. It was also among the best-equipped stations in the network boasting its own 26-meter system, but it only operated on a two-shift basis. Moreover, older stations like Quito, Santiago, and Winkfield—which could all trace their roots back to the old Minitrack days—still operated on four shifts, even well into the 1970s.

This variation in scheduling of workload had everything to do with the numbers and types of Earth science satellites that a station was called on to monitor. When one talked about the largest number of different satellites that a given station supported, Johannesburg and Orroral immediately came to mind. Each monitored 30 or more satellites in the mid-1970s. Fairbanks and Winkfield were not far behind at 24 and 22, respectively. Overall, NASA's STDN provided coverage to some 50 different satellites in the 1970s.[48] In December 1975, the GSFC made a familiar move by awarding Bendix a two year, $104 million contract—with provisions for three additional one-year extensions—to continue its role as the prime operator of the network into the decade of the 1980s and the age of the Space Shuttle.[49]

Even though American human space presence clearly saw a period of quiescence in the mid to late 1970s, science and application satellite activities continued to flourish. One area of research in which Goddard satellites and the STDN played a leading role was in tectonics—the study of the structural deformation of Earth's crust. On 4 May 1976, LAGEOS 1—the LAser GEOdynamics Satellite—was launched on a 50-year, high inclination Earth-orbit mission to study the geophysical behavior of our planet. The idea behind the mission was that long term data received from the satellite could be used to monitor the motion of Earth's tectonic plates, for example, and to measure the gravitational field and nutation (wobble) in the axis of rotation.[50]

In this activity, the exact position of the STDN on the surface of Earth was itself a piece of scientific data. Most tracking is done under the presumption that the location of the ground station is known and that tracking determines where the satellite is with respect to the ground station. In laser tectonics, the logic is reversed: the location of the satellite is known. What is desired is the exact location of the ground station. The LAGEOS satellites, covered with tiny "retroflectors," reflect laser beams transmitted from various ground stations. Covering the two-foot spherical satellite were 426 cube-corner reflectors made of fused silica glass and the heat tolerant element of germanium. By measuring the time between transmission of the beam and reception of the reflected signal from the satellite, stations on Earth can thus precisely measure the distance between themselves and the satellite. The accuracy obtained is extremely high, with distance measurements correct to within one to three centimeters, or about an inch.[51]

To return the laser data to Earth, Goddard Space Flight Center came up with the Goddard Laser Tracking Network, or GLTN, in 1975. The GLTN functioned as somewhat of a "mini network," same as the STDN but using laser instead of radio frequency signals—essentially an optical system. A laser at a GLTN station would emit a beam to the satellite which would then be reflected and returned to the station. The interval between the start of the transmission and the receipt of the return signal was recorded at the GTLN station and multiplied by the speed of light to obtain the precise distance between the station and the satellite. In this way, ever subtle changes over time in the satellite-to-station distance painted a picture of motion in Earth's tectonic plates. As a result, significant information concerning fault line movements and the dynamics of earthquakes, for example, could be deduced.

Because of tectonic science requirements, the design of GLTN stations also had to emphasize mobility. During the 1970s, most of these sites were configured into Mobile Laser Ranging Systems (MOBLAS), with each MOBLAS housing the required hardware in three instrumentation vans. By the end of the decade, the pace of laser ranging activities had picked up to where MOBLAS units had been deployed to diverse locations such as Bear Lake, Utah; Quincy and San Diego, California; and places in Australia and around the Pacific and Indian Oceans. A fixed site, meanwhile, operated on the grounds of Goddard in Greenbelt, Maryland. Like its STDN counterpart the Network Training and Test Facility, this fixed station essentially acted as a test site for development of additional MOBLAS units. To achieve even more mobility, Goddard engineers soon developed a second generation system called the Transportable Laser Ranging System, or TLRS. Instead of large instrumentation vans, the TLRS consisted of relatively small, box-like transportable units that were readily borne by trucks and aircraft. Their performance was even better than that of the MOBLAS. By 1990, these laser tracking systems had achieved astonishing ranging accuracies, down to the sub-centimeter level, a must for measuring the slow and virtually indiscernible movement of Earth's crust over time. The work continues today with the MOBLAS and the TLRS terminals still operating at Goddard. [52]

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In 1977, the NASA began a series of low altitude, atmospheric glide tests of the Space Shuttle. These Approach and Landing Tests (ALT) took place at the Dryden Flight Research Center near Edwards Air Force Base, California from February through November. The ALT was the first step in the flight qualification of the Shuttle Orbiter, verifying its flight worthiness as it glided to a landing after returning from orbit. Testing began with three ground taxi runs of the Shuttle *Enterprise* mounted atop a Boeing 747 Shuttle Carrier Aircraft—a highly modified Boeing 747-100—to determine loads,

control characteristics, steering and braking of the mated vehicles. This was followed by eight so-called "captive-flights" of the *Enterprise* (five uncrewed, the last three crewed) attached to the 747 to evaluate the structural integrity and aerodynamic performance of the mated pair in the air. Five free-flights of the Orbiter concluded the test program.

To support the ALT, Goddard set up a special mobile telemetry station in 1975 at Buckhorn Lake (a dry lake) on a hill overlooking the landing strip. Buckhorn (BUC) was a fairly simple ground station, with transportable equipment consisting of two 4.3-meter (14-foot) antennas and C-band radar, equipment in part used previously at Grand Bahama during Apollo. Trailers housed UHF air-to-ground voice and S-band telemetry equipment. After supporting the ALT and seven Shuttle orbital missions, Buckhorn was closed-out in 1983 following the STS-8 night landing. One of its 4.3-meter antennas was permanently transferred to the nearby Dryden Flight Research Facility while the other hardware was put back into the STDN equipment pool.

Three and a half years after ALT ended, the STDN tracked the first Space Shuttle into orbit. After a six year hiatus, America finally returned to space, this time ushering in a new era in space transportation with the launch of the Shuttle *Columbia* on STS-1 the morning of 12 April 1981. As Commander John W. Young and Pilot Robert L. Crippen lifted off from pad 39A at the KSC, long-time Flight Director Christopher Kraft called it the most tense moment in all his years at Mission Control. Never before had NASA flown a crew on the very first launch attempt of a new rocket. Boosted by the largest solid rocket motors ever made, the entire Shuttle stack cleared the launch tower within seconds, a surprise to those at Mission Control who remembered the painstakingly slow liftoff of the mammoth Saturn V just a decade earlier.

The 2,500 men and women of Goddard's STDN had prepared six years for the launch.[53] Even though in the eyes of the public, little activity had come from the space agency since the mid-1970s, it was quite a different story behind the scenes. Much had improved. Station equipment had been upgraded to accommodate the new multi-channel S- and Ku-band communication system of the Shuttle. Telemetry rates from tracking stations had increased to 128,000 bits per second (128 kbps) in real time versus the 14 to 21 kbps of the 1960s. Telemetry streams were transmitted to the JSC in real time on three 56 kbps circuits.[54] A key communication change was the implementation of S-band air-to-ground voice circuits in addition to UHF radio capability. The Shuttle continued to use the UHF air-to-ground voice system but the USB system of tracking, telemetry, and command developed for Apollo was now expanded to include two-way voice. This required the development and installation of equipment to digitize and multiplex voice on the command channel and de-multiplex voice from the telemetry channel. Voice was now multiplexed with commands for uplink and downlinked with telemetry. The

links were then de-multiplexed, converted to analog voice and redirected to the MCC in Houston. The more than two million circuit miles of NASCOM lines continued to be upgraded, relying on domestic and international land lines, submarine cables, commercial satellites, and microwave radio systems (not unlike relaying of cell phone signals today) to interconnect the overseas stations with the launch facility at Kennedy and the MCC in Houston.[55]

In contrast to earlier human space missions when the need for voice contacts steadily declined, the Space Shuttle program imposed a goal of 30 percent voice contact on each orbit to monitor things like critical ascent and reentry events, payload delivery tasks and on-orbit crew science activities. Studies done in the late 1970s, however, showed the STDN could provide voice support for only about 23 percent of the time. Gaps in voice coverage had to be addressed. As with earlier renditions of the network, international cooperation once again held the key.



The Dakar Station in Senegal on the western most point of the African continent looking towards the Atlantic was an ideal location to track the Shuttle's ascent into orbit on eastern launch azimuths. The arid setting was typical of west Africa as were the facilities. Clearly visible are the 4.2-meter (14-foot) USB antenna on the right and the quad-helix command antenna on the left. During Shuttle missions, DKR was staffed by about a dozen NASA contractors and Senegal workers. (Photo courtesy of Gary Schulz)

Discussions and diplomatic negotiations resulted in new stations at several important locations, each equipped with UHF air-to-ground voice systems to fill coverage gaps. As B. Harry McKeehan, former Chief of International Operations at Goddard who spearheaded many of these talks put it, "These countries gave us their full support," even in places such as Pakistan, where NASA operated a Landsat station near the industry center of Rawalpindi.[56] In June 1982, a station was added in Dakar (DKR), Senegal—the western most point of Africa—to support the Shuttle's Orbital Maneuvering System first burn (OMS-1), a critical event in the ascent to orbit timeline where a decision as whether or not to continue onto orbit or to initiate the Abort-Once-Around (AOA) sequence had to be made. DKR also provided an additional contact point for each orbit once the Shuttle was safely in orbit. In addition, Dakar served as the early Transatlantic Abort (TAL) landing site.

Also established in Africa was the Botswana Station (BOT) at Gaborone. Also called Kgale, it was added in 1981 primarily to cover the OMS-2 circularization burn. Since the Johannesburg Station was closed-out in 1975, Botswana assumed many of the functions formerly handled by the South Africa station. In the archipelagos of Seychelles, NASA called on the Air Force, tasking their 18-meter (60-foot) antenna to serve as the Indian Ocean Station (IOS) some 1,100 kilometers (700 miles) northeast of the Madagascar coastline.[57] Yarragadee (YAR) in Western Australia, was added in 1980—just prior to the launch of STS-1—to provide coverage for the Shuttle's deorbit burn and reentry.[58]

Rounding out the changes to the STDN required for Shuttle support was a 4.3-meter (14-foot) antenna atop the 1,980-meter (6,500-foot) Tula Peak (TULA) in 1979. Situated on the grounds of Holloman Air Force Base just outside the gates of White Sands Missile Range, New Mexico, TULA alleviated the approximate, 10-minute communications gap during each Shuttle orbit over the southern United States.[59]

By July of 1982, the STDN reached its zenith in terms of the number of ground stations (20 stations). For Shuttle support, it had acquired new outposts in the United States, Africa, the Indian Ocean and Australia. Network engineers had greatly enhanced its capabilities by incorporating the latest data processing and transmission innovations of the 1970s. The result was an unprecedented network with 10-fold increase in telemetry and data handling capacity over that of Apollo just a decade ago.

Despite the tide of innovation and streamlining that went into the leaner and more efficient STDN of the 1980s, NASA could not totally offset the rising cost of station operations. The network remained manpower intensive. This made operations highly susceptible to inflation, not only in the U.S., but even more so overseas, where the impact of wage escalation was even more stifling. Remote, often isolated locations were especially burdensome on cost, staffing and maintenance. Spiraling cost at certain loca-

tions—particularly Spain, Australia, Goldstone and Alaska—could no longer be ignored. The prime culprit—wage increases—had led to a 15 to 20 percent jump in operating cost at Madrid and the Australian stations, and increases of 13 and 25 percent, respectively, at Goldstone and Fairbanks. This equated to a 6 percent across the board cost increase over the entire network. While this was not huge, in the cost-conscious days following Apollo, and with double digit inflation of the late 1970s, it was enough for the space agency to begin closing down more ground stations.[60]

One of them was Rosman in North Carolina. Among the best-equipped of the original STADAN sites, Rosman had been supporting ATS-6 which was no longer operating by December of 1979, and the OAO which had completed its mission in November 1980. Although NASA pulled out of Rosman in January 1981, five years earlier it had been the target of a well-publicized (at least among the locals working there), rumored-closing. The 1976 events did not stem from technical reasons though. That year, a labor dispute arose between employees of Bendix and their company with respect to a collective bargaining agreement.

What happened was this: the station employees, who were represented by the International Brotherhood of Electrical Workers (IBEW), on 25 February of that year commenced negotiations with Bendix on their labor agreement which was soon to expire. But despite numerous meetings, an agreement could not be reached and so the union went on strike. While labor disputes were not uncommon, this strike caught the attention of the local North Carolina residents who began to feel that NASA may be thinking of closing down "their" station because of the dispute. Although the Agency really had no such intentions and (by law) had to leave IBEW and Bendix to work out their differences, state and U.S. representatives from North Carolina soon, perhaps in somewhat of a panic, got into the fray and began questioning NASA on its "true intentions."

Only after an official letter from NASA Headquarters was sent to Congressman Roy A. Taylor clarifying the Agency's position that the strike would not affect the status of the station did the rumors begin to fade. In the end, the dispute was resolved when IBEW accepted a new labor offer from BFEC which included a 15 percent wage increase—not bad considering that the average wage settlement for all major collective bargaining agreements negotiated in the U.S. during the first quarter of 1976 was just 8.8 percent.[61]

Six years later, though, dwindling pass requirements did cause NASA to really leave Rosman. But instead of just shutting it down, the DOD received authorization from Congress to assume operations of the tracking facility. As for the workers, all 119 Bendix employees assigned to the station were offered jobs elsewhere within the company. Some remained, others did not. Of the 119, 30 transferred to other Bendix locations while 34 were

retained by the DOD at Rosman. Fifty-five others declined to accept employment elsewhere and were terminated.[62]

Like Rosman, other stations were also reassigned, either within NASA or to another agency. In 1974, the NTTF became part of the operational network—Greenbelt or BLT—and expanded to take on responsibilities for NASA's IUE. This ended in 1986 when the Center decided, after the deactivation of IUE, to align all support activities at Wallops Island off the coast of Virginia. With the decision made, the 12-meter (40-foot) antenna used on IUE was given to the nearby United States Naval Academy and the 9-meter (30-foot) USB system moved to Wallops. Following this decision, this rather unique facility reverted back to its original role of serving the network as a test bed and training center.[63]

Another case in point was in Alaska after LANDSAT-3 went inoperative in 1983. On 30 September 1984, NASA operations at Fairbanks, Alaska ceased when it granted the National Oceanic and Atmospheric Administration (NOAA) a temporary-use permit to operate the station to track weather satellites. The polar orbiting Landsat was by now using TDRS-1 for support and Goddard no longer had any pressing requirements for Alaska. Being in Alaska, the station was one of the most expensive stations to operate. NASA nevertheless continued to provide operations and maintenance support to NOAA for the next four years (at a cost of $1,920,000) until a permanent transfer was finally granted by the Bureau of Land Management.[64]

In 1985, Alaska's remaining 26-meter (85-foot) antenna was transferred to the Jet Propulsion Laboratory for continued support on Nimbus and the Dynamic Explorer satellites. At the time of the station's closing, NASA had invested $12 million of capital equipment in Alaska which, after transferring to NOAA, brought to an end 26 years of NASA operations there.[65] The Agency's absence from the state, however, would be rather short-lived as it would soon return to the area, this time to conduct scientific research activities which continue to this day.

Other sites were closed out in a more permanent way. In November 1981, one of the most venerable stations in the network came to an end. During STS-2, the second flight of the Shuttle *Columbia*, the Quito Station in Ecuador was shut down as planned. Fiscal belt-tightening and steep foreign inflation rates often overpowered the international cooperation value of keeping an overseas station open. (Another consideration was the balance between international cooperation and the desire for more "U.S. territory-based solutions.") One of the original Minitrack sites, the station was located near Mount Cotopaxi, the highest active volcano in the world, 56 kilometers (35 miles) south of the Ecuadorian capital and had served faithfully as a key Southern Hemisphere station dating all the way back to 1957 and Sputnik. Bendix had operated Quito, along with the Ecuadorian Services Company

(ESCO) who provided subcontractor services, since 1961. Closing it saved NASA an estimated $4 million annually.[66]

Quito exemplified the "international value" of a NASA overseas station. It showed how the seed of a NASA station in a foreign country germinated to become a technological national resource for that country, a resource that endures to this day. Ecuador's main product is agriculture. In the late 1950s, there was an economic need for companies willing to enter into other fields such as technology and oil. In 1960, a group of Ecuadorian executives—visionaries in hindsight—led by Carlos H. "Polo" Cadena founded ESCO. BFEC soon awarded ESCO a subcontract to help operate the Quito STADAN station.

The station grew as NASA grew. It upgraded from Minitrack to a three-link station in the mid-1960s. The compliment of Ecuadorian nationals bloomed from 50 to 220. With the consolidation of STDN, it expanded from supporting only application satellites to human spaceflight support on ASTP continuing on to the Space Shuttle. This was a giant step forward for an overseas station, one that required a dynamic and joint managerial effort by Bendix and ESCO. A transition from American station staff to Ecuadorians took place and Goddard implemented its training and certification program with outstanding results. Cadena himself was a strong proponent of "station nationalization" and firmly believed that it was in the best interest of his employees.[67]

The Ecuadorian government had designated the Esceula Politecnica Nacional as NASA's cooperating agency, responsible for facilitating and monitoring the Agency's activities in that country. In the early 1970s, its Director, José Rubén Orellana, expressed dissatisfaction with NASA's integration of Ecuadorian nationals into the station staff, as provided for in the international agreement. In response, NASA brought in new station management: Charles Force was transferred from Guam while Bendix named Cliff Benson as their new Senior Manager. They quickly determined that Orellana's charges were valid and moved aggressively to remedy the situation. Over the next two years, over 50 Bendix personnel were replaced with Ecuadorian nationals. The willingness with which people like BFEC Logistics Supervisor Harry Bailey trained Fabian Mosquera as his replacement, for example—not knowing where he himself would go next—was impressive! A year or so later even Benson himself was replaced with a national: Julio Torres. Open animosity, while it did exist, was infrequent as Bendix management understood their role and made every effort to place their workers in other positions with other parts of the company. While operational performance of the station had previously been quite good, it improved even more under the new personnel, and operations costs were simultaneously reduced.[68] In 22 years, Quito provided over half a million hours (578,160 hours to be exact) of direct mission support, one of the highest in the STDN. Numerous performance awards were bestowed by both NASA and Bendix.[69]



A panoramic view of the Quito Station in 1973. The station was located at 3,650 meters (12,000 feet) elevation 69 kilometers (43 miles) south of the Equator, at the base of Mount Cotopaxi. A herd of llamas that frequented the station is slightly visible grazing just left of the 12-meter (40-foot) USB antenna. The deactivated Minitrack antenna is visible in the background between the two larger buildings. The photograph is unusual because Cotopaxi is cloud-free, and because of the rare vantage point—from the top of a communications tower along the nearby Pan American Highway that was accessible only by climbing 30 meters (100 feet) up an open ladder. (Photograph by Charles Force)

At 7:04 a.m. local time on 14 November 1981, as astronauts Joe H. Engle and Richard H. Truly passed over Mount Cotopaxi for the final time, they expressed their appreciation to the 75 station employees. Words of bittersweet thankfulness also went to the Quito crew from a host of Agency officials, including: Robert E. "Ed" Smylie, NASA's Associate Administrator for Tracking and Data Systems; John H. McElroy, Deputy Director of the Goddard Center; Richard S. Sade, NASA's Director of Networks; Mike Stevens, the Shuttle Network Manager; Walt LaFleur, Deputy Director of Networks; and Daniel A. Spintman, Chief of the Goddard Network Operations Division.[70]

The last formal agreement with Ecuador came on 4 December. On that day, the State Department authorized the U.S. Embassy to exchange



As large as the 26-meter (85-foot) antennas of the STDN were, they were dwarfed by the 70-meter (230-foot) dishes that the DSN uses to communicate with spacecraft at the outer reaches of the Solar System. This photographic rendering drives home the size of these dishes at Canberra, Goldstone and Madrid. (Photograph courtesy of NASA)

notes with Quito extending the agreement for another six months to allow NASA to perform "cleanup work" completely closing-out the station. Station equipment was transferred to Dakar, Senegal, which at the time was just being established as the Transatlantic Abort emergency landing site for the Space Shuttle. On 1 July 1982, the facility was transferred to the government of Ecuador who, in turn, assigned the CLIRSEN agency the responsibility for its operations. It has been used since to support Earth science data acquisition and regional land management and development. A number of nationals who started at the Quito Station have gone on to play an important role in the industrial development of Ecuador.[71]

Also closed during this time, with no fanfare, was the small station atop Tula Peak near Alamogordo, New Mexico. Its relatively light work-

load allowed it to be phased out and its responsibilities reassigned to other STDN sites. From a scheduling perspective, its impact was small, less than 5 percent coverage for most scientific application satellites. The loss to Space Shuttle support was even less at 3 percent and none involved mandatory or mission critical events. TULA had only been operational for less than three years but closing it would save the Agency half a million dollars a year. Just four and a half months after its closing, however, Tula Peak had to be quickly reactivated—literally overnight—to support a contingency landing of the Shuttle *Columbia* at White Sands. Due to wet ground conditions at Edwards in California, STS-3 was diverted to New Mexico (the KSC was still unavailable for Shuttle landings in 1982). Getting TULA up and running in just over 24 hours was a rather impressive feat of logistics and field engineering, a feat that once again demonstrated the "badgeless" teamwork of those who made the STDN possible.[72]

This steady phase out of the ground network continued through the 1980s. In 1981, NASA transferred ownership of its only station in England—the Winkfield Station at Berkshire—to the British, who having operated it since its establishment in 1961, continued to use it for radio research. Also realized in the big picture was the long-planned consolidation in 1985 of STDN stations at California, Australia, and Spain with their DSN counterparts. Under the reorganization, STDN capabilities were retained but now as part of the DSN. They would still be used to support the Agency's near-Earth and highly elliptical orbiting spacecraft but would be run out of the Jet Propulsion Laboratory. The thinking was that by combining the capabilities in each geographical location, more efficient use of the facilities could be realized.

First to be realigned was Goldstone. Of all the ground stations at Goldstone, only one, the Apollo Station (GDS) built in 1967, was originally part of the STDN; all others were original DSN equipment. To meet tracking requirements on the Apollo program, DSN assets were used as a wing-station, modified for USB operations and tasked to support the primary Apollo antenna. At Goldstone, the wing-station was the Pioneer Station (DSS-11), the first of the DSN sites constructed back in 1958 (it is now a National Historical Landmark). In general, a wing-station was not as well equipped as its STDN counterpart, but it provided the redundant systems (transmitters and receivers) that were needed under Apollo mission rules. This was a technically sound requirement. At lunar distances, the very narrow beamwidth of the 26-meter (85-foot) antennas (0.43°) meant that one was needed to track the Command/Service Module circling the Moon while the other was needed to focus on the Lunar Module as it made its way down and back up from the lunar surface. Under the Goldstone consolidation, Apollo GDS was reassigned to the JPL and redesignated DSS-16. A smaller 9-meter (30-foot) USB antenna was also transferred, redesignated DSS-17.[73]

An essentially parallel move was made in Spain at the Madrid Station (originally abbreviation MAD, which was changed to RID in 1984) built in

1965 at Fresnedillas 50 kilometers (30 miles) west of the capital city. The station had operated under bilateral agreements signed by the U.S. and Spain on 29 January 1964 and 11 October 1965 to establish mutual cooperation in the scientific investigation of outer space. There, the 26-meter (85-foot) STDN antenna used on Apollo was moved to Robledo by GSFC workers, placed under the auspices of JPL and redesignated DSS-66 as part of the Madrid Deep Space Communication Complex (MDSCC). Like Goldstone, MAD also had a wing-station assigned to it during Apollo. DSS-61 was just eight kilometers (five miles) away at Robledo de Chavela. It was modified for USB operations. In 1971, MDSCC became one of the first NASA tracking facilities to be turned over completely to a foreign government. Under the agreement, INTA, the Spanish National Institute of Aerospace Technology, today operates Madrid on behalf of NASA.[74]

Finally, the STDN stations half a world away in Australia were phased out. Today, mobs of wild kangaroos freely roam the abandoned grounds of Orroral Valley where one of the busiest stations once stood. ORR as it was



Abandoned in 1985, site of the Orroral Valley Station is today home to hundreds of kangaroos and their joeys in the serene valley. Shown here are remnants of where the main Operations Building used to be. (Photograph by the author)



In this picture from early April 1970, former Honeysuckle Operations Coordinator John Saxon (left) and Deputy Station Director Mike Dinn man the Ops console during pre-mission simulations for Apollo 13. Saxon holds the distinction of being the only person to have talked with an astronaut on the Moon from the Southern Hemisphere. During the Apollo 16 EVA, an earthquake knocked out the Los Angeles NASCOM node which caused Mission Control in Houston to temporarily go off the air. Since HSK was in communication with the crew at the time, Saxon chatted with Mission Commander John Young as Houston slowly got back on the air. The two agreed to share a toast if they should ever meet. They finally did—22 years later when the former mission commander visited Australia in 1994 to commemorate the 25th anniversary of the first Moon landing. ("Long Time Between Drinks," *Canberra Chronicle*, 16 July 1994. Photograph courtesy of Colin Mackellar, *www.honeysucklecreek.net/people/at_work.html*)

known, located 58 kilometers (36 miles) southwest of Canberra, was established as a STADAN facility in 1965. It was used mainly to support science and application satellites until its closure in 1985. In addition to its 26-meter (85-foot) antenna, ORR also had the Minitrack and the old Smithsonian Baker-Nunn optical cameras transferred from Island Lagoon, Woomera. In 1984, NASA shutdown the station and the USB antenna was donated to the University of Tasmania. The next year, it was moved to Mount Pleasant, east

of Hobart, Tasmania, where it stands today. The Baker-Nunn camera was also donated, but to the University of New South Wales. The remaining Minitrack control equipment was handed over to the Commonwealth Department of Territories and was used for a while by the Australian Department of Transport and Communications for monitoring small satellites.[75]

Goddard had also put in a laser ranging facility at the station in 1972 for geodetic research which was operated on behalf of NASA by the Australian Land Information Survey Group (now called Geoscience Australia) from 1975. This laser tracking facility operated at Orroral until 1998 when it was shut down and the equipment moved back to the United States. Continuing the work started at Orroral, a new laser ranging facility was established at Mount Stromlo, near the Australian Capital in 1998, wholly operated by Australia. But five years later, a devastating wildfire erupted in the hills surrounding Canberra which reached the outskirts of the city and destroyed the facility (as well as over 500 homes). A replacement facility was built in mid-2004 which continues to operate today.[76]

Just a few miles north of Orroral on Apollo Road was Honeysuckle Creek (HSK), perhaps the most historical of all the Australian sites because of its unique role on Apollo 11. In November 1981 after the second flight of the Space Shuttle, HSK closed its doors and simply faded away. Hamish Lindsay, who worked the consoles at the station, said in his book that "There were no farewells, no speeches, no parties, no wakes. All the equipment was removed, we pulled the last of the cables out, and walked out the door. During its short but glorious life, Honeysuckle Creek distinguished itself as a top station around the world in two completely different spheres as a Manned Space Flight Station and then as a Deep Space Station DSS."[77]

NASA transferred the HSK antenna to the Canberra DSN station at nearby Tidbinbilla where it has served as DSS-46 since 1983. Planned for phase-out in the coming years, the fate of the "old Honeysuckle antenna" as it is affectionately called, is nebulous. Those who worked at Honeysuckle would hate to see this piece of history simply scrapped. To this end, space enthusiasts, former station workers, and local residents in the area have banded together to form an ad hoc, private, "Save the Antenna" campaign. Their hope is that perhaps one day the historic antenna which received telemetry and video of mankind's first steps on the Moon will be restored, maybe even to stand once again at its original location in the hills of Namadgi National Park. Whether or not there will be a concerted effort by NASA or the Australian space agency CSIRO to preserve the legendary antenna in some way remains to be seen.

From a goodwill perspective, the closing of Guam was perhaps the most difficult. If ever there was a station outside of the 50 united states that could be called family, it was Guam. From the time of its ground breaking in 1966 to the later operation of the TDRSS ground node on the island, the Guamanians consistently strived for that close association with NASA,

and vice versa. An important objective in originally establishing a station on Guam was for the United States to contribute to the economic growth of the island and to serve as an educational catalyst on the territory.

In late 1988 when it became apparent that the station was going to be closed, the Guam government pleaded with NASA to keep it open. At the time, it employed 91 people, of whom almost two-thirds were hired in Guam, at an annual payroll of $3 million. In an effort to save the station, Guam Governor Joseph F. Ada formally requested that NASA Administrator James Fletcher reconsider the decision, saying "the station has lent luster to the territory of Guam and has been a great source of pride for our people."[78]

The station at Dandan, establishment of which had been such the personal campaign of Governor Guerrero, was put into caretaker status in 1989 and closed out the following year. Some equipment was left in place at the request of the State Department, who was interested in using the facilities, while the remaining equipment was transferred to the government of Guam. While many stations may have simply ceased operating without any fanfare when they were shut down, this was definitely not the case at Dandan. At the conclusion of the final pass of the Solar Maximum Mission (Solar Max) at 10:30 a.m. on 30 June 1989 Guam time, simultaneous farewells took place on the island and on the other side of the globe at NASA Headquarters. Present at the ceremony were one time Guam Station Director Charles Force; Robert Spearing, former Director of Goddard's Mission Operations and Data Systems Directorate; and a host of other NASA and contractor employees who had worked the station over the years.[79]

Going back to even before the establishment of NASA and the Minitrack days when the network was set up by the Army Corps of Engineers, the United States always tried to bring local people into the operation. The station at Santiago, Chile, was an example of where this policy worked to near perfection, even if it were at times the target of anti-American political demonstrations. The station was a remarkable example of the long-standing goodwill engendered by the networks' activities. It was eventually operated entirely by Chileans. (Even the Station Director was Chilean, working for the University of Chile under NASA contract.) Wes Bodin, the former Associate Chief for Ground Network at Goddard, explained. "This policy created a cooperative spirit with the countries NASA dealt with, created a mutual relationship. And as we phased out a station, we transferred the equipment in total to the local government. At Santiago, we transferred the entire operating entity over to the University of Chile. The University kept the Station Director and part of his crew to operate as a space tracking station."[80]

After NASA left the station, it still bought services from Chile. In the late 1980s, the university reconfigured the station to support the COSPAS-SARSAT project, a multilateral, cooperative project sponsored primarily by the United States, Canada, France, and the former Soviet Union. (COSPAS was an

acronym for the Russian phrase "Cosmicheskaya Sistyema Poiska Avariynich Sudov" meaning "Space System for the Search of Vessels in Distress" while SARSAT stood for "Search And Rescue Satellite-Aided Tracking.") The program used satellites to help search and rescue efforts by detecting signals emitted by airplanes, boats, and others in distress. Even today, the European Space Agency (ESA) and the National Space Development Agency of Japan (NASDA) continue to use such services provided by the Santiago Station.[81]

The year 1989 also saw the end of NASA operations on Ascension Island. The most isolated location in the network, Ascension ended up as one of the longest serving stations, operating without interruption for close to 25 years.[82] This streak was nearly broken, however, in 1982. From March to June of that year, the United Kingdom and Argentina engaged in a military conflict over the Falkland Islands to the south. During this brief but intense conflict, Ascension Island was used by the British for logistical support and as a result, commercial communications on and off the island were heavily disrupted. Fortunately for NASA, technical support was able to continue for the most part as the Agency maintained its own communication lines on and off the island for direct mission support. But the situation was not without its share of tense moments. Even though military action took place almost 10,000 kilometers (6,200 miles) to the south and Bendix workers on Ascension were at no time in any real danger, concerned family members back in the U.S. nevertheless had plenty of difficulty placing commercial telephone calls to their loved ones. Much of the problem was resolved when the company made available special circuits, routing telephone traffic through its headquarters in Columbia, Maryland to reach their families.[83]

Seven years later, operations at Ascension would be interrupted, this time for good. While the technical reasons to shut down Ascension Island were clear, how to close the site and what to do with it afterwards were not as obvious. Here, international cooperation with the international space community once again came to the forefront. What happened was that a series of events occurred as NASA was deciding to phase out the station, events that ended up involving three parties: the island government on Ascension, NASA, and the Europeans. Before the *Challenger* Space Shuttle accident in 1986 broke NASA's stride in constructing its Tracking and Data Relay Satellite System, or TDRSS, the space agency had planned to transfer ACN to the Air Force's Eastern Space and Missile Center (ESMC) when NASA operations ceased there in October of 1985.

To this effect, in a memorandum of understanding between GSFC Director Noel Hinners and the ESMC commander, authority and terms of the transfer were laid out in which Goddard had the responsibility to provide logistical support to ESMC for supplies and materials. Conversely, ESMC was to reimburse Goddard for contractor support provided to them during this transition period. But by 1986 when it was evident that TDRSS was going to be delayed, NASA quickly extended its agreement with British Cable and Wireless, who

provided the only way to transmit wideband data off the island. (The Air Force station, ASC, did not rely on cables but rather used high frequency radio transmissions to the Eastern Test Range as their primary communications link.)

Although the commercialization of space would not reach a full swing until the following decade, even in the 1980s it was not difficult to see that a fundamental shift in the space landscape was already taking place. This change was the movement of space from the realm of government sponsorship to commercial commodity. With the Reagan administration being a strong proponent of privatization, as the nation's space agency, this paradigm shift was not lost on NASA. In fact, the Agency had already been operating from Ascension for a few years under an agreement with the Europeans.

During this time, U.S. dealings on the island, in the words of French program officials, "has been excellent."[84] For it to work, international cooperation had to have flexibility, and on occasion, some good fortune. A case in point was the handling of coverage for Ariane's 9 November 1985 launch, which happened to coincide with Shuttle mission STS-51A. The ESA Ariane carried the GTE Spacenet 2 and the European Marecs B2 satellites while the Shuttle mission included deploying two satellites and the recovery of Westar 6 and Palapa-B2. The Shuttle launch was originally planned for 7 November and would have required Ascension Island tracking support two days later for satellite retrieval operations. It ended up, however, being delayed 24 hours causing Ascension support for the Westar and Palapa recovery to now occur on the 10th. So, because of the Shuttle delay—not an unusual occurrence—the station was now free to cover Ariane without conflict.

But according to NASA mission rules, Ascension Island was not available for Ariane support for a 48-hour period before and after a Shuttle launch, and for a similar period before and after a scheduled landing. In practice, though, flexibility in scheduling was not uncommon so as to accommodate international partners' needs. Commenting on the series of events during STS-51A in 1984, Clet Yven, a Station Chief for the French Centre National d'Etudes Spatiales CNES said:

> What we have seen in practice is that Ascension Island availability is handled on a case-by-case basis, and the periods blocked against our use depend upon the mission. They [NASA] have demonstrated excellent flexibility and have said they could free their facilities for short periods in certain cases, even when there may be general scheduling conflicts with Shuttle missions.[85]

Hence, private commercial space launch industry officials were at the time especially concerned that the station closings (not just Ascension but the others as well) would leave them without the ability to receive data from their boosters. It was logical for the United States to consider commercializing some

sites. In discussing the fate of the Ascension Station with NASA Headquarters, it was the perspective of the Office of General Counsel that "commercialization is feasible and would be consistent with other efforts to commercialize the space industry."[86] Such an arrangement could generally benefit everyone involved. In addition to the potential financial gain to the commercial operator, such an arrangement would also address these concerns, keeping the station operational while providing a continued source of revenue to benefit the local economy.

Recognizing the Europeans' need for downrange launch support out of Kourou, French Guiana, officials from NASA and CNES met at the KSC in January of 1986. On the day of the *Challenger* launch, David W. Harris, at the time Manager of Space Network Operations at GSFC, happened to be leading a contingent from Goddard to discuss with CNES these issues. Breaking the meeting to view the launch—in one of those moments indelibly etched in one's memory—the team immediately recognized the horror of the situation. Still numb from what they just witnessed, the group disbanded that morning and agreed to reconvene at a later date.[87]

Three long years would pass before the group met again on 12 January 1989 to finish their talks. NASA already had plans to transfer the facility to the Air Force ESMC at the end of the fiscal year under an existing memorandum of understanding with the DOD. Under the proposed agreement with ESA, the Europeans would in turn install their own equipment on Ascension by March of 1990, to be operated by British Cable and Wireless personnel. The station was perfect for ESA since from its spaceport at Kourou, equatorial launches of the Ariane rocket flew almost directly over the island. Therefore, ESA requested that NASA continue operating the Ascension Station just a little bit longer, on a monthly reimbursable basis, until April 1990, when their equipment would be installed and become operational.[88] Thereafter, ESA would assume full operations on its own to provide tracking services to its international customers.

An agreement was thus signed on 21 February 1989, extending NASA operations on the island on a cost reimbursable basis, one that would have ESA pay NASA $283,000 per month to keep the station open.[89] As for the facilities that ESA did not want to use, NASA was requested by the Island Administrator to restore the site to its preexisting condition. In this cleanup, the Operations Building was transferred to the Ascension government for use by the local community. All other buildings were demolished and the rubble hauled off. A significant restoration effort was the cleaning out of Devil's Ashpit which had been used as a trash pit for a quarter-century. As one can imagine, this was no easy task since The Ashpit was quite large—30 meters deep by 40 meters wide by 90 meters long (100 by 125 by 300 feet)—with sheer, fragile walls. The cleanup took a year.[90]

In this rapid succession of station closings, perhaps no other group of people was more affected on a day-to-day basis than the contractors and their

families. In 1989, BFEC for example, employed over 300 people at Ascension, Dakar, Guam, Hawaii, Santiago and Yarragardee, a good portion of whom had established families at these remote outposts (except for Ascension, which was "singles only").[91] This meant that by shutting down these stations, a few thousand people were going to be uprooted, some from the only homes they knew. Of course, there were some places where it was easier to leave than others.

Take Ascension versus Hawaii, for instance. While it may not have been all that difficult for folks to walk away from a place called Devil's Ashpit—recall that "If you can't go to the Moon, the next best place is Ascension Island!"—it was quite a different story for those who were stationed in the tropical settings of Hawaii. Located on a 25-acre site at Kokee State Park, the Hawaii Station was near Waimea Canyon on the west side of Kauai—one of the most scenic sights in the world. Since its establishment in 1961, the station had supported every U.S. human spaceflight with the exception of the first two suborbital Mercury missions. With its lush, green settings and surrounding hillside, the area is often used for motion picture and television location shots. Thus, it was not surprising that once assigned to Hawaii, one usually stayed in Hawaii.

Many of the employees at Kokee had been there for 20 years or more and had established roots there. When NASA announced that the site would be shut down at the end of the fiscal year on 30 September 1989, most of those at the station were offered positions elsewhere by Bendix. Few wanted to leave though. In the end however, with limited job opportunities on the island for skilled technicians, most took the offers and reluctantly left the island. Those who did not left the company. After 29 years of service which saw the station track John Glenn around Earth and bring back 27 astronauts from the Moon, much of the land was returned to the state of Hawaii. The station was turned over to the U.S. Navy's Pacific Missile Range. The 9-meter (30-foot) USB antenna system continued to be used for years on the Goddard Crustal Dynamics Project and is still being operated for science—tracking radio stars and studying plate tectonics by the University of Hawaii.

Finally, Botswana, Dakar and Yarragardee—the early UHF air-to-ground Shuttle voice stations—were closed in 1986, 1995, and 1991, respectively. Hardware from these stations were transferred to other STDN sites or mostly just donated to the host country. In the case of Botswana, the legacy of having hosted a "space station" in their country was preserved as NASA donated the surplus equipment to the Botswana National Museum, who made an exhibit commemorating their involvement and contribution to the success of the early Shuttle flights.[92]

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It was clear by the end of the 1980s that the era of NASA's worldwide, ground-based network had come to an end. Goddard's once sprawling STDN had been reduced by over 75 percent. Deemphasis had come a long way in just a few years. If one were to ask what the largest structures ever assembled on the face of Earth are, answers might range from the Great Wall of China to the Great Pyramids of Giza. From an infrastructure point of view, NASA's family of tracking networks—NASCOM, Minitrack, STADAN, MSFN, STDN, as well as the DSN—put together comprised one of the most wide-reaching infrastructures of the twentieth century, a true testament to the men and women who engineered it, built it, and made it work. Eventually though, technology and better access into space would supersede the need for such an extensive ground network. Instead of being tied to the surface of Earth, this new kind of network would now literally be based in space. It would change the STDN from a network using many ground stations into one using only a handful of satellites called the Tracking and Data Relay Satellite System, or TDRSS. TDRSS would enable Earth orbiting spacecraft such as the Hubble Space Telescope, the Space Shuttle and the International Space Station, to continuously communicate with control centers on the ground without an elaborate and expensive network of stations.

This fundamental change in spaceflight communications from primarily a ground-based network to a space-based network was something that NASA had in fact been working on since the early 1970s.

In other words, the revolutionary change to this new kind of network did not take place overnight.